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Symptom recognition is impaired in patients with orthostatic hypotension

Roy Freeman, MD¹, Ben MW Illigens, MD¹, Razvan Lapusca, MD², Marta Campagnolo, MD³, Ahmad R. Abuzinadah, MD⁴, Istvan Bonyhay, PhD¹, Dong-In Sinn, MD⁵, Mitchell Miglis, MD⁵, Jeffrey White, BS⁶, Christopher H Gibbons, MD¹

¹Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA

²Department of Internal Medicine/Cardiology, Heart Center Leipzig at University of Leipzig, Leipzig, Germany

³Department of Neurosciences, University of Padova, Padova, Italy

⁴King Abdulaziz University, College of Medicine and King Abdulaziz University Hospital, Internal Medicine Department, Jeddah, Saudi Arabia.

⁵Department of Neurology, Stanford Medical Center, Palo Alto, California

⁶University of Virginia School of Medicine, Charlottesville, VA

Abstract

Failure to recognize symptoms of orthostatic hypotension (OH) may result in falls, syncope and injuries. The relationship between orthostatic changes in blood pressure and symptom occurrence and severity is not known. The goal of the present study was to define the relationship between the occurrence and severity of the symptoms of orthostatic hypotension (OH) and (1) the upright systolic blood pressure (SBP) and (2) the fall in SBP after tilting in patients with OH. We prospectively studied 89 patients with OH. Reported BP values include the lowest BP in the first 3 minutes of tilt and the change in blood pressure during tilt. Subjects were queried about

Address correspondence to: Roy Freeman M.D., Autonomic and Peripheral Nerve Laboratory, Department of Neurology, Beth Israel Deaconess Medical Center, 1 Deaconess Road, Boston, MA 02215, U.S.A., Phone: (617) 632-8454, FAX: (617) 632-8454, rfreeman@bidmc.harvard.edu .

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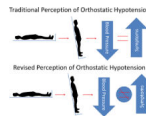
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symptoms of orthostatic intolerance while supine and during the first three minutes of tilt testing using Question 1 of the Orthostatic Hypotension Questionnaire (OHQ). Mean tilted SBP was 101.6 ± 26.1 mmHg and mean SBP fall 47.9 ± 18.1 mmHg. Mean symptom scores when upright were: lightheadedness ($2.3/10 \pm 2.7$), dizziness ($1.6/10 \pm 2.5$), and impending blackout ($0.8/10 \pm 1.9$). The majority of patients were asymptomatic or mildly symptomatic and no discrete cut-off for symptoms was observed. The magnitude of the SBP fall ($r = -0.07$, $p = \text{NS}$) and the lowest upright SBP ($r = 0.08$, $p = \text{NS}$) did not correlate with any reported symptom. These results suggest a poor relationship between the magnitude of the orthostatic BP fall, the upright orthostatic BP and symptoms. Many patients are asymptomatic despite substantial SBP falls and low orthostatic blood pressures. These findings have implications for clinical care of patients with OH and clinical trials to treat patients with OH.

Summary

This study suggests a cautious approach to the assessment of symptoms in patients with OH and the response to interventions that raise BP.

Graphical Abstract



Keywords

Orthostatic hypotension; symptoms; synucleinopathy; supine hypertension

INTRODUCTION

Classical orthostatic hypotension (OH) is defined as a sustained reduction of at least 20 mmHg of systolic blood pressure (BP) or 10 mm Hg of diastolic BP within 3 minutes of standing or head-up tilt table testing.¹ Clinical treatment of patients with OH and clinical trials of agents to treat OH are based on the assumption that patients are reliable reporters of OH symptoms. Specifically, (1) that a linear or quasi-linear relationship exists between the fall in BP and the magnitude of symptoms such that a fall in BP in patients with OH will result in a proportionate increase in symptoms and that (2) discrete BP cut-offs exist, below which the majority of patients with OH will develop symptoms.

In clinical practice, we have noted marked variations in the orthostatic BP that elicits symptoms and, that many patients with Parkinson disease and OH were unaware of clinically meaningful, reversible, cognitive impairment associated with OH.² A prior report suggested that in patients with Parkinson disease and OH, a discrete cut-off (upright MABP of <75 mmHg) separated the symptomatic from asymptomatic patients,³ while another report suggested that 33% of patients with a fall in SBP >60 mmHg were asymptomatic.⁴ These observations prompted this study, to prospectively define the relationship between the severity of OH symptoms and (1) the upright BP and (2) the fall in BP after tilting in

OH patients. We also sought to determine whether this relationship was modified by supine hypertension or the diagnosis of an alpha-synucleinopathy.

METHODS

Study Setting

This is a prospective study of individuals referred for evaluation of orthostatic hypotension at the Autonomic and Peripheral Nerve Laboratory at Beth Israel Deaconess medical Center from January 2016 through March 2018. Testing occurred in the outpatient autonomic clinical center. The protocol was approved by the Institutional Review Board of the Beth Israel Deaconess Medical Center. A complete medical history was taken from all study subjects. The data that support the findings of this study are available from the corresponding author upon reasonable request.

Autonomic Testing and Symptom Assessment

A total of 1007 individuals were studied. All individuals had a complete battery of autonomic tests including Valsalva maneuver, heart rate variability to paced breathing, 45 minute upright tilt table test and 5 minute active stand as previously described.⁵ Subjects were asked to hold all medications that could impact autonomic testing for 24 hours if possible. In brief, testing was performed in a quiet environment with continuous ECG tracing and beat-to-beat blood pressures response (Finapres, Ohmeda, Englewood, CO) for all tests. Patients rested in a supine position for at least 20 minutes prior to testing.

Blood pressures were measured non-invasively at 1-minute intervals with an automated cuff sphygmomanometer over the left brachial artery at the level of the right atrium (Colin PressMate, Colin Medical, San Antonio, TX.) in a supine position for 20 minutes (to obtain baseline stability) and during a 60° head-up tilt for 45 minutes. The average of the last 3 supine blood pressure measurement were used as the baseline blood pressure. As part of the autonomic testing protocol, we designed a questionnaire that incorporated most frequently reported symptoms of OH and included in question 1 (Q1) of the validated Orthostatic Hypotension Questionnaire (OHQ).⁶ In the supine position, prior to upright head-up tilt subjects were asked to rate symptoms of lightheadedness, dizziness, impending blackout or fainting using the questionnaire. The symptoms were rated subsequently during the first 3 minutes of upright tilt table testing when criteria for OH were met and again each time a further SBP fall of 10 mmHg was observed. A scale from 0 to 10 was used to quantify symptom intensity using a scale in which 0 designated no symptoms and 10 the worst possible symptom. The patient was classified as asymptomatic if the symptom score was 0, mildly symptomatic if the symptom score was 1–3, and symptomatic if the symptom score was >3. For binary scoring (i.e. symptoms present or absent), scores of 0–3 were considered asymptomatic and scores >3 were considered symptomatic. The symptom score for all analyses was the highest of any rated symptom on question 1 of the OHQ.

Orthostatic hypotension was defined for this study as a sustained systolic BP fall ≥ 20 mmHg¹ without recovery during the first 3 minutes compared to the pre-tilt averaged BP values measured with the cuff- sphygmomanometer.⁷ Supine hypertension was defined as a

supine SBP > 160 mmHg.⁷ Exclusion criteria included failure to meet criteria for orthostatic hypotension and failure to complete the symptom questionnaire.

Statistical Analysis

Data are presented as mean $\pm$ standard deviation. Pearson correlations were used to describe relationships between tests. Chi-square tests were used to describe differences in responses between groups. Descriptive statistics and inter-quartile differences were used to describe factors associated with symptoms. A p value <0.05 was considered significant. Bonferroni corrections were made for multiple comparisons. All statistical analysis was performed using SPSS 17 (SPSS, IBM Inc).

RESULTS

Demographics

A total of 89 participants out of the 1007 (8.8%) met criteria for inclusion in the study. The breakdown of participants is shown in the Figure 1 consort diagram. A total of 885 individuals were excluded because they did not meet criteria for OH on tilt table testing. Another 33 subjects were excluded because the symptom questionnaire was not completed during testing.

Of the included participants, 51 were male, 38 female. The mean age was 69 ± 12 years, with a mean BMI of 25.9 ± 4.7 kg/m². Complete demographic information for the participants is provided in Table 1. Supine hypertension was noted in 32/89 subjects (supine systolic BP >160 mmHg).

Specific characteristics of the study population included a pre-existing neurological diagnosis in 81% (72/89): 41 with an alpha-synucleinopathy (Parkinson disease, multiple system atrophy, Lewy body dementia or pure autonomic failure), 18 with diabetic neuropathy and 13 patients with other neuropathies. Autonomic questionnaires were completed by 73–76% of study participants. Of these, 57% reported symptoms of orthostatic intolerance with standing; 55% after a physical activity; and 25% after a meal. The demographic information for subgroups with supine hypertension and alpha-synucleinopathy is also provided in Table 1. Two-thirds (59/89) of the subjects were taking medication to treat orthostatic hypotension (midodrine (29), fludrocortisone (28), droxidopa (2), pyridostigmine (6) or more than one medication (7). More than half of the subjects (47/89) were taking anti-hypertensive medication.

Baseline Testing Characteristics

The baseline BP and heart rate characteristics for the whole group, and the individual subgroups, are reported in Table 1 and Figure 2A–2C. As a group, subjects were not symptomatic while supine. The mean score for lightheadedness was 0.6 ± 1.3 , dizziness was 0.4 ± 1.1 and mean symptom score for impending blackout or fainting was 0.2 ± 0.7 .

Upright Tilt Testing Characteristics

Hemodynamic data: The upright tilted BP and heart rate characteristics for the whole group, and the individual subgroups, are reported in Table 1. All subjects, by definition, met criteria for OH. The mean drop in systolic BP was 47.9 ± 18.1 mmHg within the first 3 minutes of upright tilt table testing. Those with supine hypertension had a greater drop in systolic BP during tilt table testing, as did the group with a diagnosed alpha-synucleinopathy. The distribution of resting supine systolic blood pressures, lowest recorded systolic blood pressures and changes in systolic BP is shown in Figure 2A–2C.

Symptoms: The symptoms of OHQ Q1 (dizziness, lightheadedness and impending blackout) ranged from 0 to 10 across the population (Figure 2D–2F). Of these symptoms, lightheadedness was the most commonly reported and had the highest reported values across the study group. Forty-three percent of patients reported no lightheadedness during tilt (OHQ score of 0); 24% of patients reported symptom scores between 1 and 3; and 33% of patients reported symptom scores of 4 or greater. Sixty-two percent of patients reported no dizziness during tilt (OHQ score of 0); 16% of patients reported symptom scores between 1 and 3; and 22% of patients reported symptom scores of 4 or greater. Eighty percent of patients reported no feeling of impending blackout during tilt; 11% of patients reported symptom scores between 1 and 3; and 9% of patients reported symptom scores of 4 or greater. Mean scores for these individual symptoms when upright are as follows lightheadedness (2.3 ± 2.7), dizziness (1.6 ± 2.5), and impending blackout (0.8 ± 1.9).

The distribution and average of the highest score of the 3 symptoms in all study participants, participants with and without alpha-synucleinopathies, and with and without supine hypertension are shown in Figure 2. Patients with supine hypertension were more likely to be asymptomatic than those without supine hypertension although when symptom score of 0 was combined with mild symptoms (i.e., a score of 0–3), there were no differences between groups (Figure 3C–3D).

The mean scores for the highest of the symptoms while upright were as follows: lightheadedness 2.3 ± 2.7 , dizziness 1.6 ± 2.5 and impending blackout or fainting 0.8 ± 1.9 .

Relationship between Symptoms and Blood Pressure Changes

By symptom group: The lowest recorded tilted systolic blood pressure, and greatest change in systolic blood pressure during tilt table testing across individuals with no symptoms (score of 0), with mild symptoms (score of 1–3) and with symptoms (score of 3–10) are shown in Figure 3A and 3B. No significant difference was observed in the lowest tilted SBP or greatest SBP fall across the 3 symptom groups (Figures 3 A and 3B).

By symptom group with supine hypertension and alpha-synucleinopathy: Those individuals with supine hypertension had a greater overall fall in SBP and higher upright SBP compared to those who were normotensive ($P < 0.01$, Table 1). Individual data points are shown in Figure 3C and 3D.

Patients with an alpha-synucleinopathy had greater supine systolic blood pressures ($P < 0.01$ vs those without an alpha-synucleinopathy, Table 1) and greater falls in systolic blood

pressure ($P < 0.01$ vs those without an alpha-synucleinopathy, Table 1). Patients with a central alpha-synucleinopathy tended to be less symptomatic at lower blood pressures than other patients although numbers were small. Individual data points are shown in Figure 3E and 3F.

By blood pressure groups: The relative proportion of patients that had no symptoms, mild symptoms or who were symptomatic were similar across the range of tilted blood pressures and blood pressure falls (Figure 4). At most blood pressures and blood pressure falls, the majority of patients were asymptomatic or only mildly symptomatic (Figure 4). This relationship held out even in those patients with supine hypertension and alpha-synucleinopathies although the number of subjects, particularly at low blood pressures, was small.

Correlations between symptoms and blood pressure: There was no correlation between the magnitude of systolic blood pressure drop and any maximal reported symptom ($R = -0.07$, $P = \text{NS}$), or any specific symptom ($R = 0.14$, $P = \text{NS}$). Similarly, when the group was subdivided into those with and those without supine hypertension, no correlation was found (with supine hypertension $R = 0.12$, $P = \text{NS}$; and without supine hypertension $R = -0.12$, $P = \text{NS}$). There was also no correlation between the lowest tilted systolic blood pressure and any maximal reported symptom, or any specific symptom in the group as a whole and those with and without OH.

DISCUSSION

OH is a highly prevalent complaint in elderly individuals and may be one of the first clinical signs in many neurodegenerative diseases, including the alpha-synucleinopathies and autonomic peripheral neuropathies.^{5, 8, 9} Patients with OH present with a wide range of symptoms⁶ that can adversely impact quality of life and in some cases may be incapacitating.^{8, 10} Prompted by observations that marked variations exist in the blood pressures that elicit symptoms in patients with OH and that some patients are unaware of symptoms of OH despite significant falls in blood pressure,^{3, 4} we sought to define the role of lowest tilted BP level and the relative changes in BP on orthostatic symptom development in patients with OH. The major findings are as follows: (1) despite low SBP on upright tilt and large SBP falls during, the majority of patients are either asymptomatic or only mildly symptomatic; (2) no discrete cut-off exists for the fall in SBP or lowest tilted SBP measure that produced symptoms in all or in the majority of participants; (3) no relationship exists between the fall in SBP or the lowest tilted SBP measure and symptom severity; and (4) of the cardinal features of OH (lightheadedness, dizziness, and impending blackout or fainting), lightheadedness was the most prevalent and usually the most severe symptom reported.

These conclusions are applicable to the group as a whole, and subgroups with and without an alpha-synucleinopathy and, with and without supine hypertension; although patients with a central alpha-synucleinopathy (MSA, PD and LBD) were more likely to be asymptomatic even at lower BPs. These data have implications for the clinical management of patients with OH and for clinical trials with interventions to treat OH. The neurobiological underpinnings of these results are not known. Possibilities include deficient sensory perception or

recognition of OH the associated decreases in cerebral perfusion, physiologic habituation, and inter-individual variations in autoregulatory mechanisms.

The functional neuroanatomy and neurophysiology for the perception of the symptoms of OH are not known. Many preclinical studies have established the neuroanatomical and neurophysiological response to hypotension and to cerebral ischemia. Specifically, changes in blood pressure activate the baroreflex which is initiated by baroreceptors – mechanoreceptors that are situated in the adventitia of the carotid sinuses and aortic arch and innervated by the glossopharyngeal and vagus nerves. This reflex restores cardiovascular homeostasis by evoking changes in peripheral vascular resistance and heart rate.^{11–13} In addition, preclinical evidence supports a cerebral ischemic response, elicited by global cerebral ischemia and characterized by arterial hypertension, increased peripheral resistance, apnea and bradycardia.^{14–16} These preclinical studies, however, do not provide the basis for the perception of hypotension and cerebral ischemia or the array of symptoms associated with these physiological stressors.

A possible explanation for the absence of symptoms in the setting of OH is habituation. In this regard, the subjective response to the stress of OH has features in common with dyspnea^{17–19} and the warning symptoms associated with hypoglycemia.²⁰ Both are symptomatic responses to a homeostatic stress and both are subject to habituation. With respect to hypoglycemia, this phenomenon, which may occur after a single episode of hypoglycemia, is known as hypoglycemia unawareness and, similarly, a previous report used the term hypotension unawareness to describe the lack of symptoms in some patients with severe OH.⁴ While knowledge exists about the peripheral and central anatomical substrates for the warning symptoms for hypoglycemia²⁰ and dyspnea^{17–19} and the mechanisms that underlie habituation in these conditions, little is known about the anatomical basis for the conscious recognition of OH and nothing is known about potential mechanisms of habituation. The observation that patients with alpha-synucleinopathies and OH were more likely to be asymptomatic with SBP less than 100 mmHg than those with other causes of OH may provide some insight into the underlying neuroanatomy. In these patients, attenuation of the subjective response to hypotension may be secondary to the degeneration of those structures responsible for sensing impaired cerebral perfusion or could represent a form of anosognosia, specifically attenuation of interoception,^{21, 22} due to the neurodegenerative process. However, the number of subjects with central alpha-synucleinopathies and SBP less than 100 mmHg in this study cohort is small and a definitive conclusion will require a larger number of subjects.

Autoregulation could also underlie the absence of symptoms in some patients with OH. Autoregulatory mechanisms maintain a constant cerebral blood flow between mean arterial pressures of 60 mmHg-160 mmHg, and many of our study participants had orthostatic blood pressures within that range. Most,^{23–27} although not all²⁸ studies, suggest that cerebral autoregulation is maintained in autonomic failure, likely due to non-autonomic factors that include endothelial, myogenic and metabolic mechanisms. However, some study participants, many of whom were asymptomatic, had orthostatic blood pressures below the autoregulatory threshold. It is possible that non-autonomic autoregulatory mechanisms are enhanced in patients with autonomic failure,²⁵ although currently, no evidence exists to

support a role for enhanced autoregulation in the absence or reduction of symptoms in these study participants. It is also not known why some patients are symptomatic with orthostatic blood pressures within the range of autoregulation.

The discordance between orthostatic BP and symptoms in this study emphasizes the need for special attention when taking care of patients with OH. Falls, trauma-induced fractures and head injury are a common cause of morbidity and hospitalization^{10, 29} and a significant contributor to cost of medical care in patients with OH.³⁰ Our findings may contribute to understanding these observations. The brain's metabolic reserve to compensate for ischemia and hypoxia is limited - unconsciousness occurs 6–8 seconds after the cessation of cerebral perfusion^{31, 32} and lack of symptoms is not likely to provide protection for this limited metabolic reserve. Thus, patients with OH are at high risk for syncope and falls, with and without prodromal symptoms. Physicians and other caretakers involved in the care of OH patients should not be reassured by the absence of symptoms. On the contrary, the absence of warning protective symptoms may place patients at greater risk for falls and subsequent injuries.

Our results differed from a prior report that found a discrete cut-off (a MABP of < 75mmHg) for the appearance of symptoms in patients with orthostatic hypotension. Methodologic differences may contribute to the differences between studies. These differences include that only Parkinson's disease patients were studied in the earlier report while we studied patients with all causes of orthostatic hypotension; and, that more than 50% of patients were tested by standing in the earlier report whereas all of our patients were tested on the tilt-table.

Limitations

A limitation of this study is that symptoms and symptom scores were actively requested, thus, our results may have over-estimated the recognition of symptoms. Also, because symptoms were elicited in the laboratory, and OH was evoked by tilt-table testing and not active standing, these data may not be applicable to real world settings. In addition, our results may not apply to acute or non-neurogenic OH; most patients included in the study had pre-existing neurological disorders causing OH. Also, although pressor medications were discontinued 24 hours prior to study, we cannot exclude the possibility that medication use may have influenced test results. Finally, we report our results in relationship to the SBP based on the report that approximately 95% of patients with classical orthostatic hypotension can be identified by systolic criterion alone³³ and, as noted in that report, the main determinants of brain blood flow are the absolute level of arterial pressure and the pulse pressure.³² Thus, an isolated fall in diastolic pressure is not likely to induce significant hypoperfusion of the brain³² and the evidence that symptoms of orthostatic intolerance such as dizziness or (pre-)syncope are strongly dependent on SBP and not on DBP decline.³⁴ Reanalysis of data using DBP and MABP produces similar results.

While the cross-sectional data presented in this manuscript raise concerns that an increase in BP in patients with symptomatic OH will not result in a proportionate improvement in symptoms, this may differ in individual patients. Despite these limitations, the data suggest a cautious approach is warranted when treating and studying patients with OH, and that

symptoms and BP measurements are not necessarily interchangeable in the assessment of the patient with OH and the response to therapies.

PERSPECTIVES

We report that many patients are asymptomatic despite substantial SBP falls and low orthostatic blood pressures. Further, a linear relationship does not exist between the symptoms of OH and both the magnitude of the orthostatic BP fall and the upright orthostatic BP. These findings have important implications for registration clinical trials, because patient-reported orthostatic symptoms are the primary efficacy endpoint for the U.S. Food and Drug Administration approval of drugs to treat orthostatic hypotension. The present data highlight the importance of measuring orthostatic vital signs and suggest a cautious approach is warranted to the assessment of symptoms in patients with OH and the response to interventions that raise BP.

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Novelty and Significance:**What is New?**

- Many patients with chronic diseases that result in orthostatic hypotension (OH) are asymptomatic despite substantial systolic blood pressure falls and low orthostatic blood pressures.

What is Relevant?

- The inability of many patients with chronic disease to detect symptoms of OH has serious implications for clinical care of patients where screening for orthostatic symptoms may not be a reliable means of detecting OH.
- The inability of patients to detect orthostatic symptoms is highly relevant in any clinical trials of OH where symptom scores are considered an outcome measure.

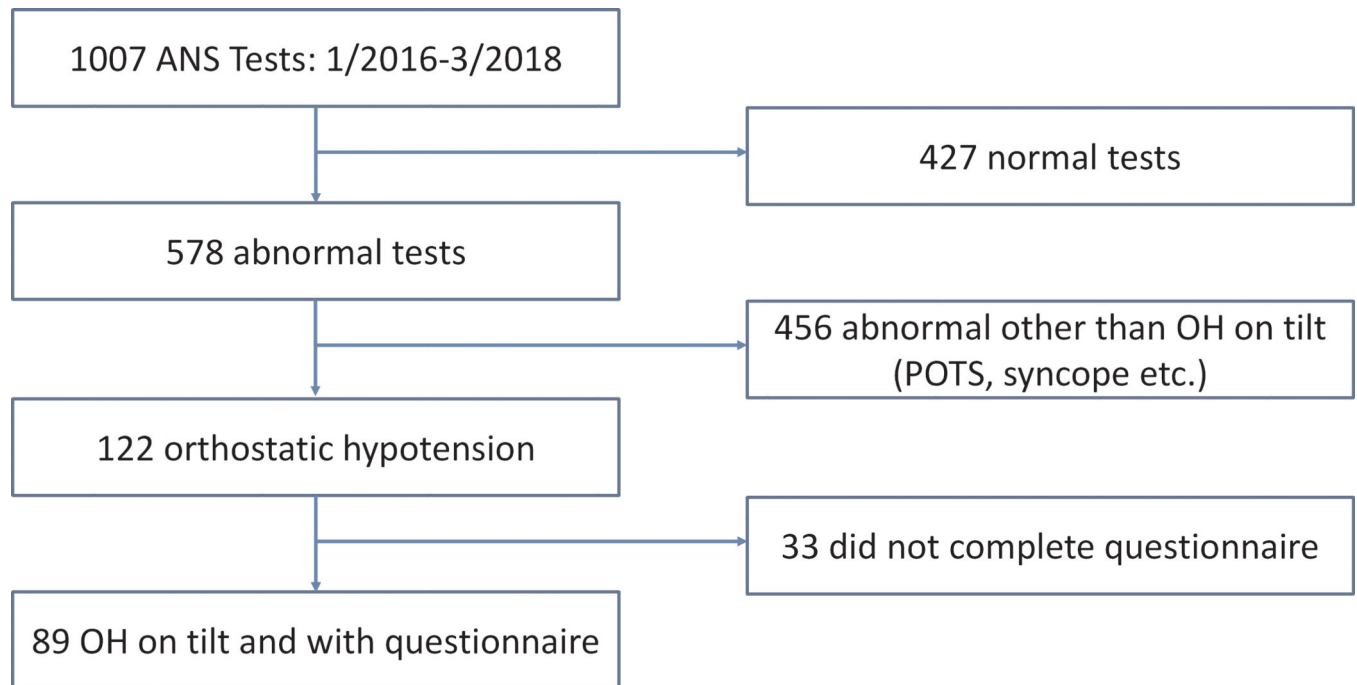


Figure 1:

Consort diagram of patients studied. 1007 patients with autonomic nervous system (ANS) testing were reviewed. Individuals with normal results, or other disorders (such as postural tachycardia syndrome - POTS) that did not meet criteria for orthostatic hypotension (OH) were excluded. Thirty-three individuals were excluded because the symptom questionnaire was not delivered during the tilt-table test.

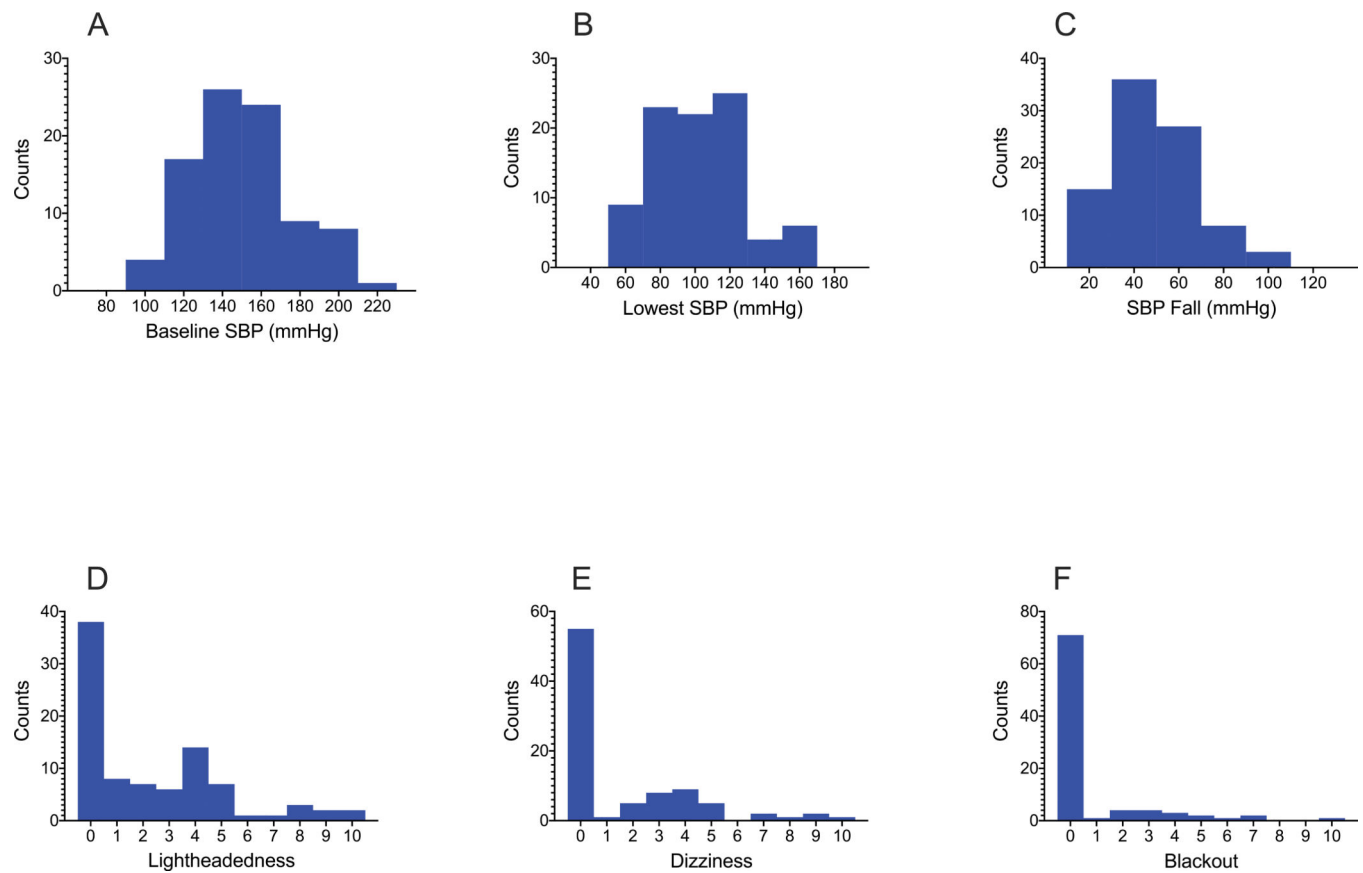


Figure 2:

The frequency distribution of the baseline SBP (A), and the lowest tilted SBP (B) and magnitude of SBP fall during the first 3 minutes of tilt (C). The frequency distribution of the symptoms of question 1 of the Orthostatic Hypotension Questionnaire (OHQ) is shown in 2D-2F. Symptoms were rated on a 0–10 scale where 0 denotes “none” and 10 denotes the “worst possible” symptom. The highest of these symptoms was chosen as the representative symptom for the analyses. In D, symptoms of lightheadedness are reported, in E symptoms of dizziness and in F, symptoms of blacking out or fainting.

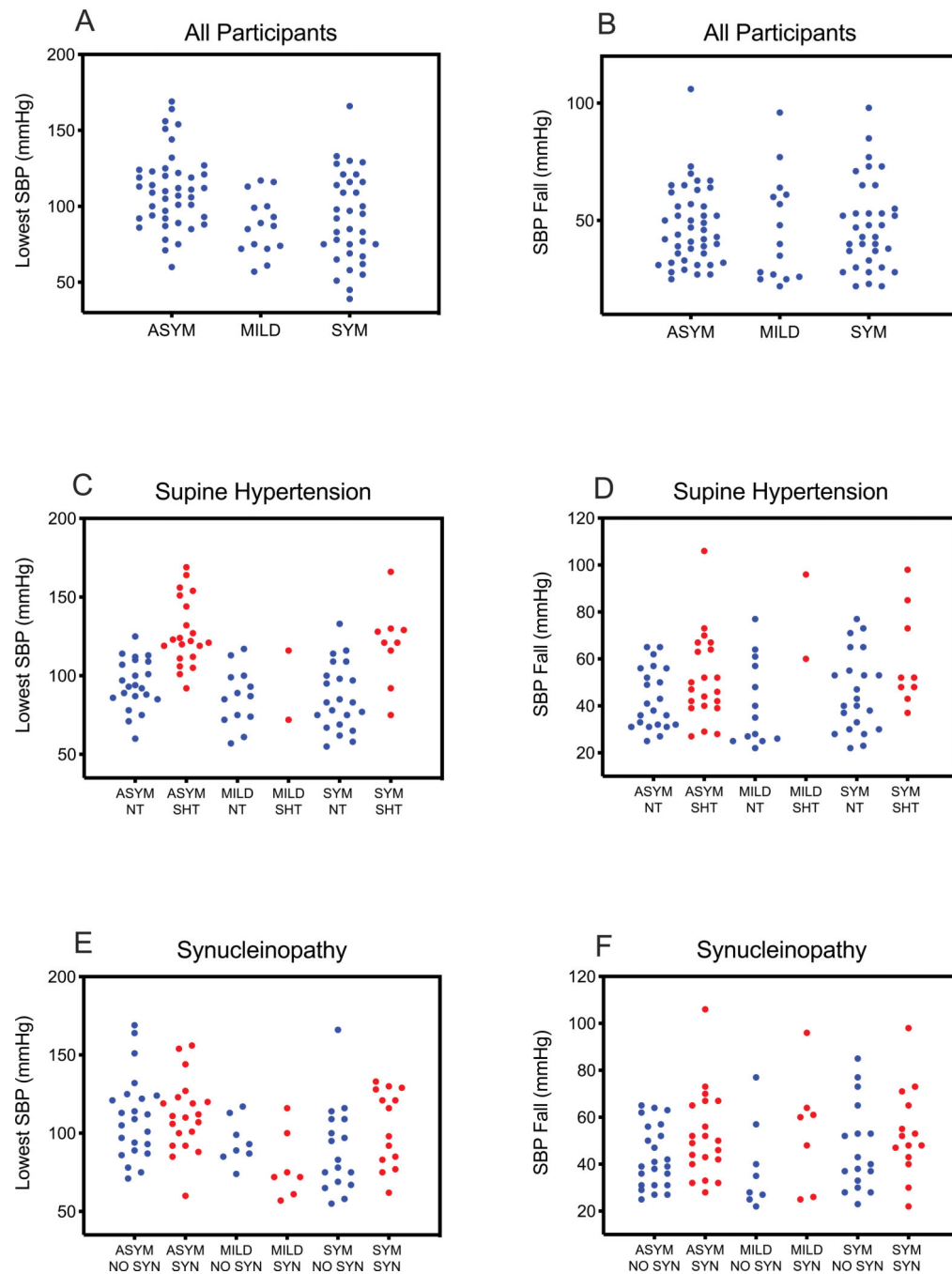


Figure 3:

The relationship between the lowest tilted SBP (3A) and the SBP fall (3B) during the first 3 minutes of tilt and highest rated symptom on question 1 of the OHQ. ASYM=Asymptomatic (Score of 0); MILD=Mild symptoms (Score of 1–3); SYM=Symptomatic (Score of 4–10). Figure 3C-3F: The relationship between the lowest tilted SBP during the first 3 minutes of tilt and highest rated symptom on question 1 of the OHQ, subdivided by the absence (blue circle) or presence (red circle) of supine hypertension (3C & 3D) and the absence (blue circle) or presence (red circle) of an alpha-synucleinopathy (3E & 3F).

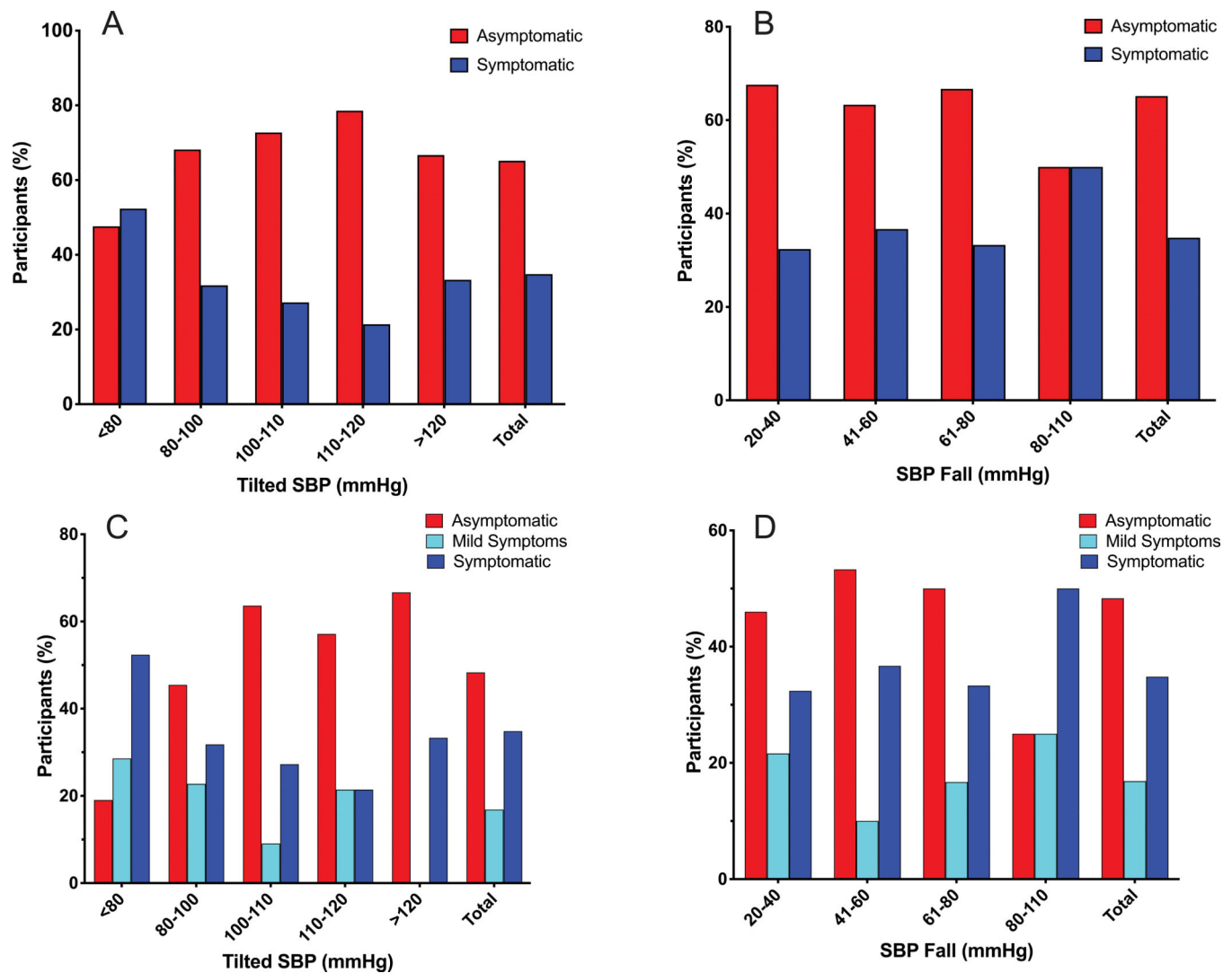


Figure 4.

The percent of patients that are asymptomatic (symptoms rated 0/10), or have any symptoms at all (rated 1–10/10) across a range of upright tilted blood pressures (4A). In 4C, the same information is shown across the fall in upright blood pressures. In 4B, the patients are subdivided into asymptomatic (0/10) or mildly symptomatic (rating 1–3) or symptomatic (rating 4–10) across a range of upright tilted blood pressures. In 4D, the same information is shown across the fall in upright tilted blood pressures.

Table 1:

Demographic Information

	All participants (n=89)	With supine hypertension (n=32)	Normotensive (n=57)	Alpha-synucleinopathy (n=41)	No alpha-synucleinopathy (n=48)
Gender M:F	51:38	19:13	32:25	30:11	21:27
Age (years)	69.4 ± 11.9	72.5 ± 8.7	67.7 ± 13.2	73.0 ± 8.9	66.3 ± 13.3
BMI (kg/m ²)	25.9 ± 4.7	26.1 ± 5.6	25.8 ± 4.1	26.4 ± 4.2	25.4 ± 5.1
SBP baseline (mmHg)	149.6 ± 26.5	179.6 ± 14.9	133.2 ± 15.1*	155.7 ± 27.2	144.3 ± 25.0†
DBP baseline (mmHg)	78.7 ± 12.5	88.2 ± 11.3	73.4 ± 9.9*	82.4 ± 13.2	75.5 ± 11.1
MAP baseline (mmHg)	102 ± 15.7	118.4 ± 10.0	93.4 ± 10.1*	106.9 ± 16.1	98.5 ± 14.3†
HR baseline (bpm)	70 ± 12.2	70.5 ± 12.7	69.9 ± 12.0	69.3 ± 10.6	70.7 ± 13.4
Lowest SBP – first 3' (mmHg)	101.6 ± 26.1	123.0 ± 23.7	89.6 ± 18.7*	101.9 ± 25.8	101.4 ± 26.6
Lowest DBP – first 3' (mmHg)	57.4 ± 15.8	66.3 ± 14.5	52.4 ± 14.3*	57.5 ± 16.5	57.2 ± 15.2
Lowest MAP – first 3' (mmHg)	76.6 ± 20.6	91.3 ± 19.1	68.3 ± 16.5*	76.8 ± 21.8	76.3 ± 19.8
Associated HR (bpm)	78.4 ± 15.0	75.8 ± 13.3	79.9 ± 15.7	76.4 ± 13.0	80.1 ± 16.4
Highest HR – first 3'	80 ± 15.6	77.4 ± 13.7	81.7 ± 16.5	78.4 ± 13.6	81.6 ± 17.2
SBP fall (mmHg)	47.9 ± 18.1	55.5 ± 20.0	43.6 ± 15.6*	53.8 ± 19.5	42.8 ± 15.3†
DBP fall (mmHg)	21.3 ± 13.3	21.9 ± 15.1	20.9 ± 12.4	24.9 ± 13.5	18.2 ± 12.5
HR increase	9.8 ± 8.6	6.6 ± 6.0	11.6 ± 9.3*	9.1 ± 7.2	10.4 ± 9.6